shape of space—you might be hard pressed to answer. Even in your groggy state, you know that Einstein showed space to be kind of like Silly Putty and so, in principle, it can take on practically any shape. How, then, can you possibly answer your interrogator's question? We live on a small planet orbiting an average star on the outskirts of a galaxy that is but one of hundreds of billions dispersed throughout space, so how in the world can you be expected to know anything at all about the shape of the entire universe? Well, as the fog of sleep begins to lift, you gradually realize that the power of symmetry once again comes to the rescue.

If you take account of scientists' widely held belief that, over large-scale averages, all locations and all directions in the universe are symmetrically related to one another, then you're well on your way to answering the interrogator's question. The reason is that almost all shapes *fail* to meet this symmetry criterion, because one part or region of the shape fundamentally differs from another. A pear bulges significantly at the bottom but less so at the top; an egg is flatter in the middle but pointier at its ends. These shapes, although exhibiting some degree of symmetry, do not possess complete symmetry. By ruling out such shapes, and limiting yourself only to those in which every region and direction is like every other, you are able to narrow down the possibilities fantastically.

We've already encountered one shape that fits the bill. The balloon's spherical shape was the key ingredient in establishing the symmetry between all the Lincolns on its swelling surface, and so the three-dimensional version of this shape, the so-called *three-sphere,* is one candidate for the shape of space. But this is not the only shape that yields complete symmetry. Continuing to reason with the more easily visualized two-dimensional models, imagine an *infinitely* wide and *infinitely* long rubber sheet—one that is completely uncurved—with evenly spaced pennies glued to its surface. As the entire sheet expands, there once again is complete spatial symmetry and complete consistency with Hubble's discovery: every Lincoln sees every other Lincoln rush away with a speed proportional to its distance, as in Figure 8.4. Hence, a three-dimensional version of this shape, like an infinite expanding cube of transparent rubber with galaxies evenly sprinkled throughout its interior, is another possible shape for space. (If you prefer culinary metaphors, think of an infinitely large version of the poppy seed muffin